

```
import java.util.Scanner;
```

```
public class Mergesort {
```

```
    public static void mergesort(int arr[], int si, int ei)
```

```
    {
        if (si >= ei) {
            return;
        }
    }
```

```
    int mid = si + (ei - si) / 2;
```

```
    mergesort(arr, si, mid);
```

```
    mergesort(arr, mid + 1, ei);
```

```
    merge(arr, si, mid, ei);
```

```
    public static void merge(int arr[], int si,
                             int mid, int ei)
```

```
    {
        int temp[] = new int [ei - si + 1];
```

```
        int i = si;
```

```
        int j = mid + 1;
```

```
        int k = 0;
```

```
        while (i <= mid && j <= ei) {
```

```
            if (arr[i] < arr[j]) {
```

```
                temp[k] = arr[i];
```

```
                i++;
```

```
            }
```

```
            else {
```

```
                temp[k] = arr[j];
```

```
                j++;
```

```
            }
```

```
        k++;
    }
```

```
    while (i <= mid) {
```

```
        temp[k++] = arr[i++];
```

```
    }
```

```
    for (k = 0; i = si; k < temp.length; k++, i++)
```

```
    {
        arr[i] = temp[k];
    }
```

```
    }
```

```
    public static void printArr(int arr[]) {
```

```
        for (int i = 0; i < arr.length; i++) {
```

```
            System.out.print(arr[i] + " ");
```

```
        }
```

```
        System.out.println();
```

```
    }
    public static void main(String[] args)
```

```
    {
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        int arr[] = new int [n];
```

```
        for (int i = 0; i < n; i++)
```

```
        {
            arr[i] = sc.nextInt();
        }
```

```
    }
    mergesort(arr, 0, arr.length - 1);
```

```
    printArr(arr);
```

```
    sc.close();
```

```
    }
```

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\
6
```

```
450 1200 800 650 300 1500
```

```
300 450 650 800 1200 1500
```

```
PS C:\Users\Aashish\OneDrive\Desktop\java> []
```

import java.util.Scanner;

public class OrderProcessAnalysis {

public static void mergesort (int arr[], int si, int ei)

{
if (si >= ei) {
return;

}
int mid = si + (ei - si) / 2;

mergesort (arr, si, mid);
mergesort (arr, mid + 1, ei);
merge (arr, si, mid, ei);

}

public static void merge (int arr[], int si, int mid,
int ei) {

int temp[] = new int [ei - si + 1];

int i = si;

int j = mid + 1;

int k = 0;

while (i <= mid & j <= ei) {

if (arr[i] < arr[j]) {

temp[k] = arr[i];

i++;

} else {

temp[k] = arr[j];

j++;

}
k++;

}

while (i <= mid) {

temp[k++] = arr[i++];

}

while (j <= ei) {

temp[k++] = arr[j++];

}

for (k = 0, i = si; k < temp.length; k++, i++) {

arr[i] = temp[k];

}

public static void main (String[] args) {

Scanner sc = new Scanner (System.in);

int n = sc.nextInt();

int times[] = new int[n];

for (int i = 0; i < n; i++) {

times[i] = sc.nextInt();

}

mergesort (times, 0, n - 1);

for (int i = 0; i < n; i++) {

System.out.print (times[i] + " ");

}

System.out.println();


```

double median;
if(n%2 == 1) {
    median = times[n/2];
} else {
    median = (times[n/2 - 1] + times[n/2]) / 2.0;
}
if(median == (long) median) {
    System.out.println("Median:" + (long) median);
} else {
    System.out.println("Median:" + median);
}

int count = 0;
for(int i = 0; i < n; i++) {
    if(times[i] > median) {
        count++;
    }
}
System.out.println("Orders Above Median:" + count);

int diff = times[n-1] - times[0];
System.out.println("Difference:" + diff);
sc.close();
}

```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS

7

12 25 18 30 15 22 40

12 15 18 22 25 30 40

Median: 22

Orders Above Median: 3

Difference: 28

PS C:\Users\Aashish\OneDrive\Desktop\java>

```

import java.util.Scanner;

public class Quicksort {

    public static void quicksort (int arr[], int si, int ei)
    {
        if (si >= ei)
        {
            return;
        }
        int pidx = partition(arr, si, ei);
        quicksort(arr, si, pidx-1);
        quicksort(arr, pidx+1, ei);
    }

    public static int partition (int arr[], int si, int ei)
    {
        int pivot = arr[ei];
        int i = si-1;
        for (int j=0; j<ei; j++)
        {
            if (arr[j] <= pivot)
            {
                i++;
                int temp = arr[j];
                arr[j] = arr[i];
                arr[i] = temp;
            }
        }
        i++;
    }
}

```

```

int temp = pivot;
arr[ei] = arr[i];
arr[i] = temp;
return i;
}

public static void main (String args[])
{
    Scanner sc = new Scanner(System.in);

    int n = sc.nextInt();

    int arr[] = new int[n];
    for (int i=0; i<n; i++) {
        arr[i] = sc.nextInt();
    }
    quicksort(arr, 0, arr.length-1);
    printArr(arr);
    public static void printArr (int arr[])
    {
        for (int i=0; i<arr.length; i++)
        {
            System.out.print (arr[i] + " ");
        }
        System.out.println();
    }
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\O
6
120 45 78 200 60 95
45 60 78 95 120 200
PS C:\Users\Aashish\OneDrive\Desktop\java>

```



```

import java.util.Scanner;

public class TradeAnalytics {

    public static void quicksort (int arr[], int si, int ei)
    {
        if (si > ei)
        {
            return;
        }
        int pidx = partition (arr, si, ei);
        quicksort (arr, si, pidx-1);
        quicksort (arr, pidx+1, ei);
    }

    public static int partition (int arr[], int si, int ei)
    {
        int pivot = arr[ei];
        int i = si-1;
        for (int j = si; j < ei; j++)
        {
            if (arr[j] >= pivot)
            {
                i++;
                int temp = arr[j];
                arr[j] = arr[i];
                arr[i] = temp;
            }
        }
        i++;
        int temp = pivot;
    }

```

```

    arr[ei] = arr[i];
    arr[i] = temp;
    return i;
}

public static void printArr (int arr[])
{
    for (int i = 0; i < arr.length; i++)
    {
        System.out.print (arr[i] + " ");
    }
    System.out.println();
}

public static void main (String args[])
{
    Scanner sc = new Scanner (System.in);
    int n = sc.nextInt();
    int arr[] = new int[n];
    int sum = 0;
    for (int i = 0; i < n; i++)
    {
        arr[i] = sc.nextInt();
        sum = sum + arr[i];
    }
    quicksort (arr, 0, arr.length-1);
    printArr (arr);
    int topCount = 5;
    if (n < 5) topCount = n;
    System.out.print ("Top 5: ");
    int topSum = 0;

```

```

    for (int i = 0; i < topCount; i++)
    {
        System.out.print (arr[i] + " ");
        topSum = topSum + arr[i];
    }
    System.out.println();
    int topAvg = topSum / topCount;
    System.out.println ("Average of Top 5: " + topAvg);
    int overallAvg = sum / n;
    int countAbove = 0;
    for (int i = 0; i < n; i++)
    {
        if (arr[i] > overallAvg)
        {
            countAbove++;
        }
    }
    System.out.println ("Values Above Overall Average: "
        + countAbove);
}
}

```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\O
8
500 1200 700 2000 900 1500 3000 2500
3000 2500 2000 1500 1200 900 700 500
Top 5: 3000 2500 2000 1500 1200
Average of Top 5: 2040
Values Above Overall Average: 3
PS C:\Users\Aashish\OneDrive\Desktop\java>

```